بيانك
Bayanuk

Database Schema

PostgreSQL schema for the Bayanuk MVP: tables, relationships, indexes, and data retention.

Product	Bayanuk (بيانك), AI review of job offers and employment contracts
Owner	JX Tech Studio
Version	1.2 (updated for the API-based AI stack)
Date	October 2026

Changes in v1.2

This version follows the AI Stack Decision v2.0: analysis and chat use the Gemini API and embeddings use Voyage AI. It replaces v1.1 (self-hosted models). Rows changed since v1.0 are marked v1.2. Apply these changes as a new migration; never edit migrations that are already merged.

Where	Change
3.5 document_chunks	`embedding` is `vector(1024)` (voyage-4). `embedding_model` is `voyage-4`.
3.6 analyses	`provider` is `gemini`; `model` holds the exact model ID; `cost_usd` is computed from the price table.
3.4 document_pages	`vision` marks pages read by Gemini after Tesseract OCR fails.
3.13 usage_events	`cost_usd` feeds the daily AI budget of $2.
8 Points to confirm	Embedding model and vector dimension are decided.

1. Summary

This document defines the PostgreSQL schema for the Bayanuk MVP. It is written for the Laravel backend (migrations and Eloquent models) and is the reference for the API and frontend fields. The schema supports guest uploads, the processing pipeline, structured extraction with source references, attention points, document chat with citations, offer comparison (MVP+), usage tracking, and verifiable deletion.

Decision	Choice	Reason
Database	PostgreSQL 16+ with the pgvector extension	One database for relational data and embeddings; no extra vector store to run.
Primary keys	UUID (ordered, Laravel HasUuids)	Safe to expose in URLs; IDs cannot be guessed or counted.
Files	Not stored in the database	Files live in private S3-compatible storage (RustFS); the database keeps only the storage key.
Flexible data	jsonb for the full extraction result	The AI output schema will evolve; normalized key fields are copied to their own table for queries.
Embeddings	voyage-4 (Voyage AI), vector(1024)	Multilingual, low cost; within the HNSW 2,000-dimension limit. v1.2
Enums	varchar + CHECK constraint, mirrored by PHP enums	Easier to change than native PostgreSQL enum types.
Time	timestamptz, stored in UTC	Displayed in the user's time zone by the frontend.
Deletion	Hard delete with cascade for user content	Privacy promise: "delete" means gone. Only a content-free deletion log is kept.

2. Entity relationships

```
users 1 ──── * documents                (documents.user_id, nullable for guests)
documents 1 ── * document_pages         (extracted text per page)
documents 1 ── * document_chunks        (RAG chunks + embeddings)
documents 1 ── * analyses               (one per run; latest is shown)
analyses  1 ── * extracted_terms        (normalized key fields with source)
analyses  1 ── * findings               (attention points)
analyses  1 ── * analysis_feedback      (thumbs up/down, field corrections)
documents 1 ── * chat_sessions 1 ── * chat_messages
users 1 ──── * comparisons ── 2 documents (document_a_id, document_b_id)   [MVP+]
users / guests ── * usage_events        (limits and AI cost)
deletion_logs                           (no foreign keys; survives deletion)
```

Guests are identified by a random guest token kept in the browser. Only its SHA-256 hash is stored (`documents.guest_token_hash`). When a guest signs in, their documents are claimed: `user_id` is set and the hash is cleared.

3. Tables

3.1 users

Column	Type	Constraints	Notes
`id`	`uuid`	PK	
`name`	`varchar(120)`	nullable	Optional; magic-link users may not provide it.
`email`	`varchar(255)`	unique, not null	Stored lower-case.
`email_verified_at`	`timestamptz`	nullable	Set on first successful magic-link or Google login.
`google_id`	`varchar(64)`	unique, nullable	From Google sign-in.
`locale`	`varchar(2)`	default 'ar', CHECK in (ar, en)	Interface language and default summary language.
`terms_version`	`varchar(20)`	not null	Version of terms and privacy policy accepted.
`terms_accepted_at`	`timestamptz`	not null	
`ai_training_opt_in`	`boolean`	default false	Must stay false unless the user explicitly opts in. Not used in MVP.
`keep_documents`	`boolean`	default false	If true, documents are not auto-deleted after 30 days.
`created_at / updated_at`	`timestamptz`		
`deleted_at`	`timestamptz`	nullable	Account deletion request; a job then hard-deletes all user data.

3.2 login_tokens

Magic-link sign-in. One-time, short-lived tokens.

Column	Type	Constraints	Notes
id	uuid	PK	
email	varchar(255)	not null, index	
token_hash	char(64)	unique	SHA-256 of the token sent by email. The raw token is never stored.
expires_at	timestamptz	not null	15 minutes after creation.
used_at	timestamptz	nullable	A used token cannot be used again.
ip_hash	char(64)	nullable	For rate limiting and abuse checks.
created_at	timestamptz		

3.3 documents

One row per uploaded file. The central table of the product.

Column	Type	Constraints	Notes
id	uuid	PK	Used in the result URL.
user_id	uuid	FK users, nullable, on delete cascade	Null for guests.
guest_token_hash	char(64)	nullable, index	Owner for guest uploads.
original_filename	varchar(255)	not null	Shown to the owner only.
mime_type	varchar(100)	not null	Validated by content, not extension.
file_size	integer	CHECK <= 10485760	Bytes. 10 MB limit.
sha256	char(64)	not null	Detects duplicate uploads by the same owner.
storage_key	varchar(255)	unique, not null	Object key in RustFS. Never the original filename.
page_count	smallint	nullable, CHECK <= 30	
language	varchar(5)	nullable	ar, en, or mixed.
document_type	varchar(30)	nullable	job_offer, employment_contract, other.
status	varchar(20)	not null, index	See enum in section 4.
status_detail	varchar(255)	nullable	User-safe reason when failed or rejected.

Column	Type	Constraints	Notes
is_scanned	boolean	default false	True if OCR was needed.
processing_started_at	timestamptz	nullable	
processing_finished_at	timestamptz	nullable	Used for the time-to-result metric.
expires_at	timestamptz	not null, index	Auto-deletion time (24 h guest, 30 days user).
created_at / updated_at	timestamptz		

3.4 document_pages

Column	Type	Constraints	Notes
id	uuid	PK	
document_id	uuid	FK documents, cascade	
page_number	smallint	not null	1-based. Unique with document_id.
text	text	not null	Cleaned text with personal identifiers masked.
char_count	integer	not null	
extraction_method	varchar(20)	not null	text_layer, ocr, or vision (page image read by Gemini when OCR fails). v1.2

3.5 document_chunks

Retrieval units for the chat. A chunk never crosses a page boundary, so every answer can cite one page.

Column	Type	Constraints	Notes
id	uuid	PK	
document_id	uuid	FK documents, cascade, index	
page_number	smallint	not null	
chunk_index	smallint	not null	Order inside the document.
content	text	not null	
token_count	smallint	not null	
embedding	vector(1024)	not null	voyage-4 (Voyage AI), 1024 dimensions. v1.2

Column	Type	Constraints	Notes
embedding_model	varchar(60)	not null	voyage-4 in MVP. Allows re-embedding if the model changes. v1.2

3.6 analyses

One row per analysis run. A document normally has one; re-runs (new prompt version, model change, retry, user request) add rows. The latest successful row is shown.

Column	Type	Constraints	Notes
id	uuid	PK	
document_id	uuid	FK documents, cascade, index	
status	varchar(20)	not null	pending, running, succeeded, failed.
provider	varchar(30)	not null	gemini in MVP. Values: gemini, plus any backup provider added later. v1.2
model	varchar(80)	not null	Exact model identifier used, for example the Gemini 3.8 Flash model ID. v1.2
prompt_version	varchar(20)	not null	Prompts are versioned in the repository.
schema_version	varchar(20)	not null	Version of the extraction JSON schema.
result	jsonb	nullable	Full validated extraction output.
summary	text	nullable	Plain-language summary.
summary_language	varchar(2)	nullable	ar or en.
input_tokens / output_tokens	integer	default 0	Summed across all AI calls of this run (all stages). v1.2
cost_usd	numeric(10,5)	default 0	Computed from the price table in config. Feeds the daily AI budget check. v1.2
duration_ms	integer	nullable	
error_code	varchar(50)	nullable	Internal code; never raw provider errors to users.
created_at / updated_at	timestamptz		

3.7 extracted_terms

Normalized copy of the key fields in analyses.result, one row per field. Used by the result cards, comparison, rules engine, and analytics without parsing JSON.

Column	Type	Constraints	Notes
id	uuid	PK	
analysis_id	uuid	FK analyses, cascade, index	
key	varchar(60)	not null	Dotted key, for example compensation.basic_salary or notice.employee_days.
value_text	text	nullable	Text value as normalized.
value_number	numeric(14,2)	nullable	Numeric value (Arabic numerals converted).
unit	varchar(20)	nullable	SAR, days, months, hours_per_week.
found	boolean	not null	False means "not found in document".
page_number	smallint	nullable	Source page.
source_quote	varchar(500)	nullable	Short quote; verified to exist in the page text.
confidence	numeric(3,2)	nullable	0.00 to 1.00 from the model; informational in MVP.
corrected_by_user	boolean	default false	MVP+: user correction of a field.

Unique index on `(analysis_id, key)`.

3.8 findings

Attention points shown to the user.

Column	Type	Constraints	Notes
id	uuid	PK	
analysis_id	uuid	FK analyses, cascade, index	
category	varchar(40)	not null	non_compete, long_notice, long_probation, penalties, salary_mismatch, missing_info, one_sided_change, working_time, other.
severity	varchar(10)	not null	info, attention, high.

Column	Type	Constraints	Notes
source	varchar(10)	not null	rule (deterministic) or ai (clause detection).
rule_id	varchar(40)	nullable	Config key of the rule that fired, for testing.
title	varchar(160)	not null	Short heading in the summary language.
explanation	text	not null	"Why it matters", never a legal verdict.
page_number	smallint	nullable	
source_quote	varchar(500)	nullable	
sort_order	smallint	default 0	High first.

3.9 chat_sessions

Column	Type	Constraints	Notes
id	uuid	PK	
document_id	uuid	FK documents, cascade	One session per document in MVP.
user_id	uuid	FK users, nullable	Null for guests.
question_count	smallint	default 0	Enforces the 3 (guest) / 20 (user) limit.
created_at	timestamptz		

3.10 chat_messages

Column	Type	Constraints	Notes
id	uuid	PK	
chat_session_id	uuid	FK chat_sessions, cascade, index	
role	varchar(10)	CHECK in (user, assistant)	
content	text	not null	
citations	jsonb	default '[]'	[{"page":7,"chunk_id":"...","quote":"..."}]
answer_found	boolean	nullable	False when the answer is not in the document.
input_tokens / output_tokens	integer	default 0	
created_at	timestamptz		

3.11 comparisons (MVP+)

Column	Type	Constraints	Notes
id	uuid	PK	
user_id	uuid	FK users, cascade	Comparison requires sign-in.
document_a_id	uuid	FK documents, cascade	
document_b_id	uuid	FK documents, cascade	CHECK a <> b.
result	jsonb	nullable	Row-by-row values from extracted_terms.
summary	text	nullable	AI summary of the difference.
created_at	timestamptz		

3.12 analysis_feedback

Column	Type	Constraints	Notes
id	uuid	PK	
analysis_id	uuid	FK analyses, cascade	
rating	smallint	CHECK in (-1, 1)	Thumbs down or up.
field_key	varchar(60)	nullable	Which field was wrong, if given.
comment	varchar(1000)	nullable	
created_at	timestamptz		

3.13 usage_events

Append-only record used for limits (uploads per day), the daily AI budget, and cost reporting. Holds no document content.

Column	Type	Constraints	Notes
id	bigint	PK (identity)	High volume; uuid not needed.
user_id	uuid	nullable, index	No FK, so cost history survives account deletion.
guest_token_hash	char(64)	nullable, index	
ip_hash	char(64)	nullable	For guest rate limits.
event	varchar(30)	not null	upload, analysis, chat_question, comparison.

Column	Type	Constraints	Notes
input_tokens / output_tokens	integer	default 0	
cost_usd	numeric(10,5)	default 0	Real cost per event. The daily AI budget ($2) is checked against the sum for the current day. v1.2
created_at	timestamptz	index	

3.14 deletion_logs

Proof that deletion happened, without keeping what was deleted.

Column	Type	Constraints	Notes
id	bigint	PK (identity)	
document_id	uuid	not null (no FK)	The ID only.
reason	varchar(20)	not null	user_request, retention_expired, account_deleted.
file_deleted	boolean	not null	Storage object removal confirmed.
deleted_at	timestamptz	not null	

3.15 Laravel default tables

Kept as generated by the framework: `sessions`, `cache`, `jobs`, `job_batches`, `failed_jobs`, `personal_access_tokens` (only if token auth is used instead of SPA cookies). The default `password_reset_tokens` table is not needed because there are no passwords.

4. Enumerations

Enum	Values
documents.status	uploaded, scanning, extracting, classifying, analyzing, indexing, ready, failed, rejected
documents.document_type	job_offer, employment_contract, other
documents.language	ar, en, mixed
analyses.status	pending, running, succeeded, failed
findings.severity	info, attention, high
findings.source	rule, ai
document_pages.extraction_method	text_layer, ocr, vision v1.2: vision = Gemini fallback
deletion_logs.reason	user_request, retention_expired, account_deleted

> **Status flow:** uploaded → scanning → extracting → classifying → analyzing → indexing → ready. Any stage can go to failed. Classifying can end in rejected (document_type = other). The frontend polls the status and maps each value to a friendly progress label.

5. Indexes

Index	Purpose
`documents (user_id, created_at desc)`	"My documents" list.
`documents (guest_token_hash)`	Guest access and claim on sign-in.
`documents (expires_at)`	Hourly retention job.
`documents (user_id, sha256)`	Duplicate upload check.
`analyses (document_id, created_at desc)`	Latest analysis lookup.
`extracted_terms (analysis_id, key) unique`	Field lookup and comparison.
`document_chunks USING hnsw (embedding vector_cosine_ops)`	Vector search on vector(1024); always filtered by document_id first. v1.2
`usage_events (user_id, created_at), (guest_token_hash, created_at)`	Daily limits.

6. Example: analyses.result (jsonb)

```
{
  "schema_version": "1.0",
  "document_type": "employment_contract",
  "language": "ar",
  "employer_name":    { "value": "Company X", "page": 1, "quote": "..." },
  "contract": {
    "type":            { "value": "fixed_term", "page": 1, "quote": "..." },
    "duration_months": { "value": 12, "page": 1, "quote": "..." }
  },
  "probation_days":    { "value": 90, "page": 2, "quote": "..." },
  "notice": {
    "employee_days":   { "value": 30, "page": 7, "quote": "..." },
    "employer_days":   { "value": null, "found": false }
  },
  "compensation": {
    "currency": "SAR",
    "basic_salary":    { "value": 8000, "page": 3, "quote": "..." },
    "housing":         { "value": 1500, "page": 3, "quote": "..." },
    "transportation":  { "value": 500, "page": 3, "quote": "..." },
    "other": [],
    "total_monthly":   { "value": 10000, "page": 3, "quote": "...", "computed": false }
  },
  "annual_leave_days": { "value": 21, "page": 5, "quote": "..." },
  "non_compete": { "exists": true, "duration_months": 12, "page": 9, "quote": "..." }
}
```

7. Retention and deletion rules

- **User request:** deleting a document removes the storage object first, then the database row (cascade removes pages, chunks, analyses, terms, findings, chats). A row is added to deletion_logs.
- **Retention job:** runs hourly; deletes every document with `expires_at < now()` the same way.
- **Account deletion:** deletes all of the user's documents, then the user row. usage_events rows are kept without content for cost reporting.
- **Backups:** database backups expire after 7 days, so deleted data does not live forever in backups. This is stated in the privacy policy.
- **Failed storage deletion:** the job retries; the database row is only removed after the storage object is confirmed gone.
- **AI providers:** only the text each task needs is sent to Gemini and Voyage AI, on paid plans. Files, results, and embeddings are stored on our server only. v1.2

8. Points to confirm before implementation

Point	Status
Embedding model and its vector dimension	**Decided:** voyage-4, vector(1024). See the AI Stack Decision v2.0. v1.2
Retention defaults: 24 hours for guests and 30 days for users	To confirm
Chat limits: 3 questions (guest) and 20 (user) per document	To confirm
Whether comparisons ship in the MVP or right after	To confirm (the table can be created now either way)
Migration policy	Apply the schema as one initial migration set; later changes go through new migrations only, never by editing old ones.